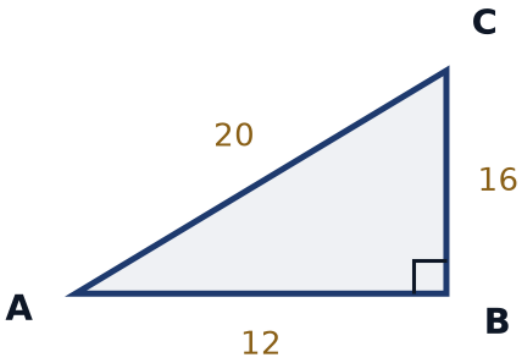
Lines, Triangles & Polygons

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

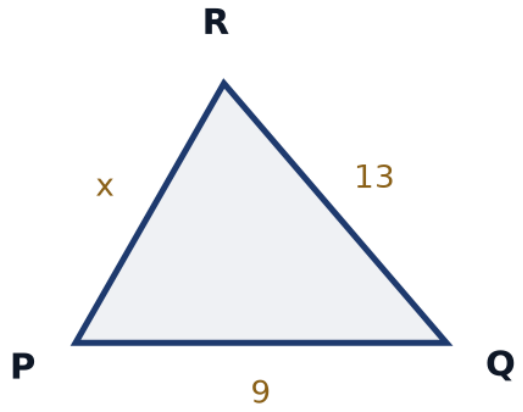
1 **[Foundational] • Target: Similar triangles and proportional reasoning**

Triangle ABC shown below has $AB = 12$, $BC = 16$, and $CA = 20$. It is similar to triangle DEF, where $DE = 15$ corresponds to AB. Find EF and FD.



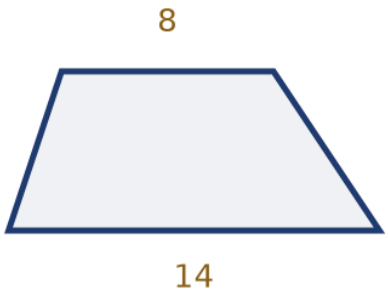
2 **[Foundational] • Target: The Law of Cosines**

In triangle PQR shown below, $PQ = 9$, $QR = 13$, and $\angle Q = 50^\circ$. Find $PR = x$, to the nearest hundredth.



3 **[Foundational] • Target: Area of polygons**

The trapezoid below has parallel sides of length 14 and 8, and height 6. Find its area.



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4 [Foundational] • Target: Parallel lines and transversals

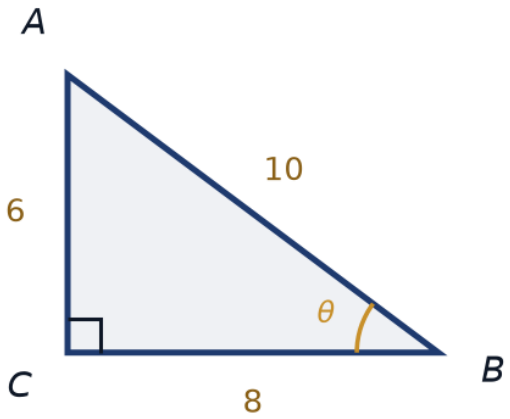
Two parallel lines are cut by a transversal, forming an angle of 105° on one line. Find the corresponding angle, the alternate interior angle, and the co-interior (same-side interior) angle on the other line.

5 [Foundational] • Target: Parallel lines and transversals

Two angles formed by a transversal crossing parallel lines are alternate exterior angles, so they are equal. If one measures $3x + 10$ and the other measures $5x - 30$, find x and the measure of the angle.

6 [Foundational] • Target: Parallel lines and transversals

Two parallel lines are cut by a transversal. One interior angle is $4x + 15$, and its corresponding angle on the other line is $6x - 25$. Find x and the angle measure.



7 [Foundational] • Target: Triangle angle sum and the exterior angle theorem

The angles of a triangle are in the ratio $3 : 5 : 7$. Find each angle.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Triangle angle sum and the exterior angle theorem

In triangle ABC, $\angle A = 52^\circ$ and the exterior angle at C is 130°. Find $\angle B$.

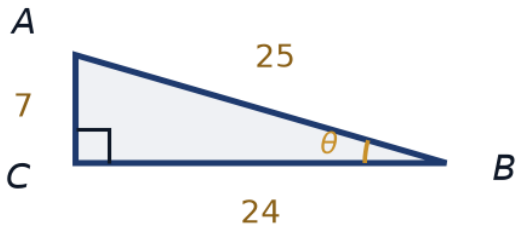
9 [Intermediate] • Target: Triangle angle sum and the exterior angle theorem

An exterior angle of a triangle measures 115°, and one remote interior angle is 50°. Find the other remote interior angle and the third (adjacent) interior angle.

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10 [Intermediate] • Target: Triangle congruence criteria

State which congruence criterion (SSS, SAS, ASA, AAS, or RHS) proves triangles PQR and XYZ congruent, given $PQ = XY$, $\angle Q = \angle Y$, and $QR = YZ$.



11 [Intermediate] • Target: Triangle congruence criteria

Triangle DEF has $DE = 9$, $EF = 12$, $\angle E = 90^\circ$, and triangle GHI has $GH = 9$, $HI = 12$, $\angle H = 90^\circ$. Are the triangles congruent? By which criterion?

12 [Intermediate] • Target: Triangle congruence criteria

Two triangles have all three pairs of corresponding sides equal in length. Which congruence criterion applies, and are the triangles necessarily congruent?

13 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle $ABC \sim$ triangle DEF, with $AB = 6$, $BC = 8$, and $DE = 9$. Find EF.

14 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle JKL has $JK = 15$ and $KL = 21$, and is similar to triangle MNO with scale factor $\frac{2}{3}$ (triangle MNO is smaller), where MN corresponds to JK. Given $MN = 10$, verify the scale factor is consistent, then find NO.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

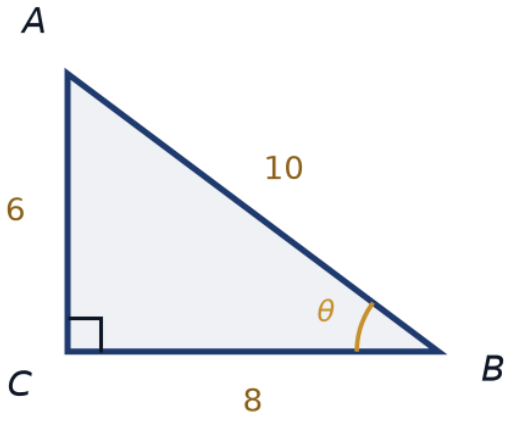
15 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 5.01 Trigonometry

A triangle has sides $a = 7$, $b = 8$, $c = 10$. Use the Law of Cosines to find angle C (opposite side c), to the nearest tenth of a degree.

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16 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 6.01 Circles & Coordinate Geometry

A triangle has sides $a = 6$, $b = 6$, and included angle $C = 80^\circ$. Find side c, to the nearest hundredth, and state what type of triangle this is based on the two given equal sides.



17 [Challenging] • Target: Heron's formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

Find the area of a triangle with sides 13, 14, 15 using Heron's formula.

18 [Challenging] • Target: Heron's formula • Cross-Lesson: 8.01 Statistics & Probability

A triangular plot of land has sides 9, 10, 17. Find its area using Heron's formula.

19 [Challenging] • Target: Heron's formula • Cross-Lesson: 1.01 Advanced Algebra

A triangular sail has sides 5 m, 12 m, and 13 m. Find its area using Heron's formula, and verify using the right-triangle area formula (since 5-12-13 is a Pythagorean triple).

20 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Find the sum of the interior angles of a regular decagon (10 sides), and the measure of each interior angle.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Parallel lines and transversals

The angles of a triangle are in the ratio $2 : 1 : 3$. Find each angle.

a. $30, 64, 86$

b. $30, 60, 90$

c. $32, 60, 88$

d. $90, 60, 30$

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22 [Foundational] • Target: Triangle angle sum and the exterior angle theorem

The angles of a triangle are in the ratio $9 : 3 : 3$. Find each angle.

a. $38, 36, 106$

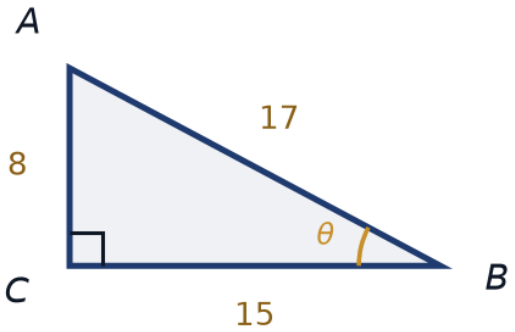
b. $36, 40, 104$

c. $36, 36, 108$

d. $108, 36, 36$

23 [Foundational] • Target: Triangle congruence criteria

The angles of a triangle are in the ratio $1 : 5 : 6$. Find each angle.



a. $90, 75, 15$

b. $15, 79, 86$

c. $15, 75, 90$

d. $17, 75, 88$

24 [Foundational] • Target: Similar triangles and proportional reasoning

The angles of a triangle are in the ratio $4 : 5 : 6$. Find each angle.

a. $50, 60, 70$

b. $48, 64, 68$

c. $48, 60, 72$

d. $72, 60, 48$

25 [Foundational] • Target: The Law of Cosines

The angles of a triangle are in the ratio $3 : 3 : 9$. Find each angle.

a. $108, 36, 36$

b. $38, 36, 106$

c. $36, 40, 104$

d. $36, 36, 108$

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26 [Foundational] • Target: Heron's formula

The angles of a triangle are in the ratio $9:5:6$. Find each angle.

a. $45, 54, 81$

b. $81, 54, 45$

c. $47, 54, 79$

d. $45, 58, 77$

27 [Foundational] • Target: Interior and exterior angles of polygons

The angles of a triangle are in the ratio $5:6:1$. Find each angle.

a. $15, 75, 90$

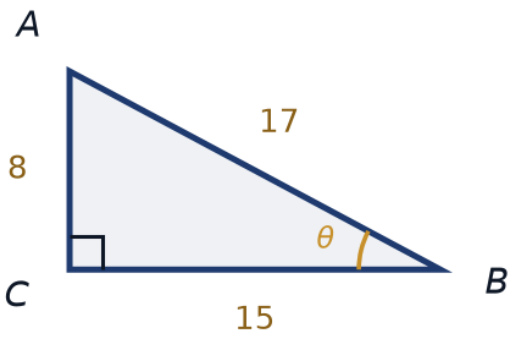
b. $17, 75, 88$

c. $90, 75, 15$

d. $15, 79, 86$

28 [Foundational] • Target: Area of polygons

An exterior angle of a triangle measures 82°, and one remote interior angle is 30°. Find the other remote interior angle.



a. 52°

b. 57°

c. 47°

d. 30°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a surveyor's triangular lots

An exterior angle of a triangle measures 76°, and one remote interior angle is 44°. Find the other remote interior angle.

a. 27°

b. 44°

c. 37°

d. 32°

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30 [Intermediate] • Target: Coordinate Geometry of Lines

An exterior angle of a triangle measures 56°, and one remote interior angle is 28°. Find the other remote interior angle.

a. 28°

b. 31°

c. 23°

d. 33°

31 [Intermediate] • Target: Triangle Theorems & Inequalities

An exterior angle of a triangle measures 54°, and one remote interior angle is 23°. Find the other remote interior angle.

a. 23°

b. 36°

c. 31°

d. 26°

32 [Intermediate] • Target: Polygons & Interior/Exterior Angles

An exterior angle of a triangle measures 88°, and one remote interior angle is 22°. Find the other remote interior angle.

a. 22°

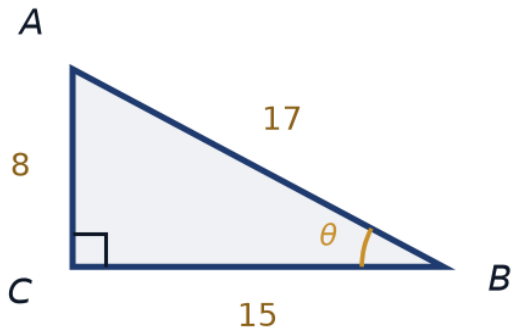
b. 71°

c. 61°

d. 66°

33 [Intermediate] • Target: Geometric Transformations & Dilations

An exterior angle of a triangle measures 86°, and one remote interior angle is 54°. Find the other remote interior angle.



a. 32°

b. 54°

c. 27°

d. 37°

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34 [Intermediate] • Target: Parallel lines and transversals

An exterior angle of a triangle measures 47°, and one remote interior angle is 25°. Find the other remote interior angle.

a. 25°

b. 22°

c. 27°

d. 17°

35 [Intermediate] • Target: Triangle angle sum and the exterior angle theorem

A triangle has sides $a = 12$, $b = 10$, and included angle $C = 40^\circ$. Find side c, to the nearest hundredth.

a. 8.26

b. 8.76

c. 7.26

d. 7.76

36 [Intermediate] • Target: Triangle congruence criteria

A triangle has sides $a = 10$, $b = 6$, and included angle $C = 70^\circ$. Find side c, to the nearest hundredth.

a. 10.74

b. 9.74

c. 9.24

d. 10.24

37 [Intermediate] • Target: Similar triangles and proportional reasoning

A triangular plot of land has sides 24, 45, 51. Find its area using Heron's formula.

a. 542

b. 60

c. 540

d. 538

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 8.01 Statistics & Probability

A triangular plot of land has sides 36, 105, 111. Find its area using Heron's formula.

a. 1890

b. 1888

c. 126

d. 1892

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39 [Challenging] • Target: Heron's formula • Cross-Lesson: 1.01 Advanced Algebra

Find the sum of the interior angles of a regular polygon with 72 sides, and the measure of each interior angle.

a. $12780, 170$

b. $12600, 175$

c. $12600, 180$

d. $12420, 175$

40 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Find the sum of the interior angles of a regular polygon with 30 sides, and the measure of each interior angle.

a. $5220, 163$

b. $5040, 168$

c. $4860, 168$

d. $5040, 173$

41 [Challenging] • Target: Coordinate Geometry of Lines • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the perpendicular distance from the point $(3, -2)$ to the line $5x - 12y + 8 = 0$.

a. 2.5

b. 3.6

c. 4.0

d. 4.5

42 [Challenging] • Target: Coordinate Geometry of Lines • Cross-Lesson: 5.01 Trigonometry

What is the acute angle of intersection between the lines $y = 2x + 1$ and $y = -3x + 4$?

a. 45°

b. 60°

c. 30°

d. 75°

43 [Challenging] • Target: Triangle Theorems & Inequalities • Cross-Lesson: 6.01 Circles & Coordinate Geometry

The side lengths of a triangle are $5, 12,$ and 13. What is the radius of the incircle (inradius)?

a. 1

b. 2

c. 3

d. 4

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44 [Challenging] • Target: Polygons & Interior/Exterior Angles • Cross-Lesson: 7.01 Solid Geometry & Similarity

How many diagonals does a regular decagon (10-sided polygon) have?

a. 25

b. 35

c. 45

d. 70

45 [Challenging] • Target: Geometric Transformations & Dilations • Cross-Lesson: 8.01 Statistics & Probability

A triangle with an area of 14 cm^2 is dilated by a scale factor of $k = 3$. What is the area of the image triangle?

a. 42 cm^2

b. 84 cm^2

c. 126 cm^2

d. 168 cm^2

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